

PAPER CODE: BEM-C202
SEMESTER EXAMINATION (MAY-2026)
CLASS: B.Tech. SEMESTER: II
SUBJECT: ENGINEERING MATHEMATICS II

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL																								
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 x 6 = 30 Marks)																											
Q-1	Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$	CO1	L1																								
Q-2	Solve $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 3y = 2xe^{3x}$	CO1	L5																								
Q-3	Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \sin x \cos 2y$	CO2	L2																								
Q-4	Solve $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = x + y$	CO2	L5																								
Q-5	Prove that $4J_n''(x) = J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)$	CO3	LL3																								
Q-6	Show that $\int_{-1}^1 P_m(x)P_n(x)dx = 0$, $m \neq n$	CO3	L4																								
Q-7	Find the Fourier series for $f(x) = e^{-x}$, in the interval $(0, 2\pi)$.	CO4	L4																								
Q-8	Find the Fourier series for $f(x) = \sin ax$, in the interval $(-\pi, \pi)$.	CO4	L5																								
Q-9	In a certain factory producing cycle tyres, there is a small chance of 1 in 500 tyres to be defective. The tyres are supplied in <u>lots</u> of 10. Using Poisson distribution, calculate the approximate number of lots containing no defective, one defective and two defective tyres respectively, in a <u>consignment</u> of 10,000 lots.	CO5	L3																								
Q-10	The following marks have been obtained by a class of students in Hindi,<table><tr><td>Paper I</td><td>80</td><td>45</td><td>55</td><td>56</td><td>58</td><td>60</td><td>65</td><td>68</td><td>70</td><td>75</td><td>85</td></tr><tr><td>Paper II</td><td>81</td><td>56</td><td>50</td><td>48</td><td>60</td><td>62</td><td>64</td><td>65</td><td>70</td><td>74</td><td>90</td></tr></table>Compute the coefficient of correlation for the above data.	Paper I	80	45	55	56	58	60	65	68	70	75	85	Paper II	81	56	50	48	60	62	64	65	70	74	90	CO5	L4
Paper I	80	45	55	56	58	60	65	68	70	75	85																
Paper II	81	56	50	48	60	62	64	65	70	74	90																
SECTION-B (LONG ANSWER TYPE QUESTIONS)		CO	BL																								

Instructions: Answer any four questions in detail. Each question carries 10 marks.																															
										(4 x 10 = Marks)																					
Q-1	Solve $\frac{d^3y}{dx^3} - 2\frac{dy}{dx} + 4y = e^x \cos x$									CO1	L1																				
Q-2	Solve by changing the independent variable $x^6 \frac{d^2y}{dx^2} + 3x^5 \frac{dy}{dx} + a^2y = \frac{1}{x^2}$									CO1	L2																				
Q-3	Solve by the method of separation of variables $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$									CO2	L5																				
Q-4	Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial y} - z = \cos(x + 2y)$									CO2	L3																				
Q-5	Solve in series the given differential equation $y_2 - xy_1 + x^2y = 0$									CO3	L4																				
Q-6	Find the Fourier series for $f(x) = x^2 - 2, -2 \leq x \leq 2$									CO4	L4																				
Q-7	State and prove Generating function for Bessel's function.									CO5	L3																				
Q-8	Calculate the first four moments of the following distribution about the mean and hence find β_1 & β_2 .									CO5	L5																				
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr></table>										x	0	1	2	3	4	5	6	7	8	f	1	8	28	56	70	56	28	8	1		
x	0	1	2	3	4	5	6	7	8																						
f	1	8	28	56	70	56	28	8	1																						
